Deep Learning Laboratory - Assignment 4

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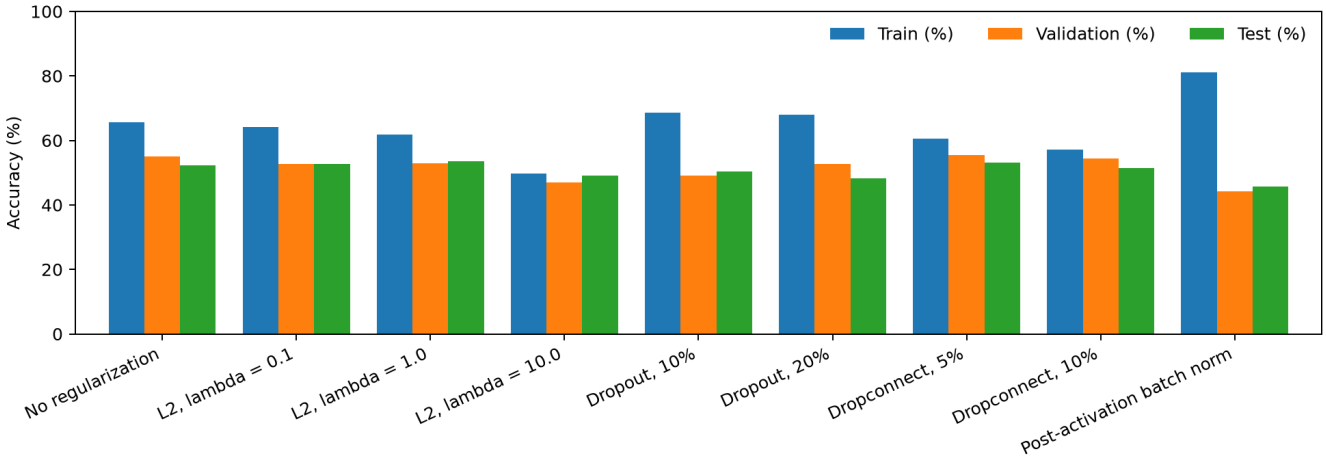
Setup

Network: $36 \to 30 \to 20 \to 5$; ReLU hidden layers; softmax output; five mini-batches; Adam. All studies start from the same weights.

Training stops after five consecutive epochs with $|E_t - E_{t-1}| < 10^{-3}$, or at 300 epochs. For L2, the training loss is $\mathrm{CE} + \lambda \sum \|W\|^2/(2N)$.

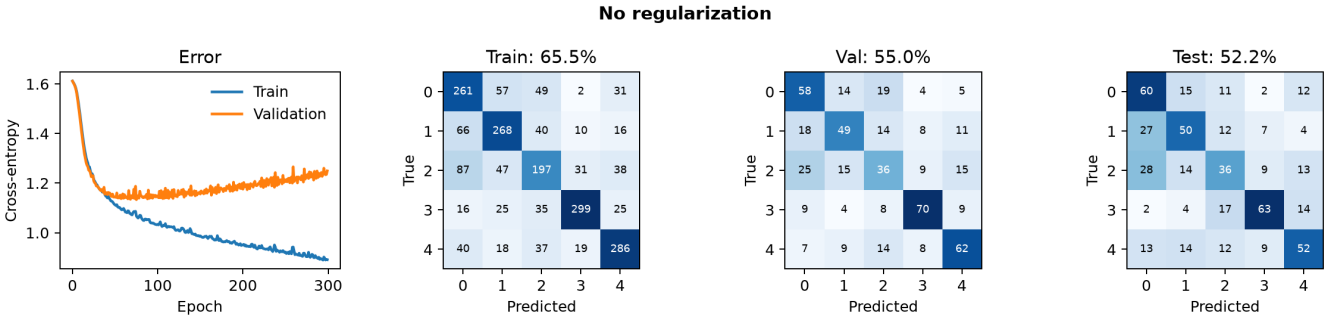
Accuracy

Method	Epochs	Train (%)	Val. (%)	Test (%)
No regularization	300	65.55	55.00	52.20
L2, $\lambda = 0.1$	300	64.20	52.80	52.60
L2, $\lambda = 1.0$	300	61.90	53.00	**53.60**
L2, $\lambda = 10.0$	151	49.70	47.00	49.00
Dropout, 10%	300	68.60	49.00	50.40
Dropout, 20%	300	67.90	52.80	48.20
Dropconnect, 5%	300	60.45	**55.40**	53.20
Dropconnect, 10%	300	57.15	54.40	51.40
Post-activation batch norm	300	81.15	44.20	45.80

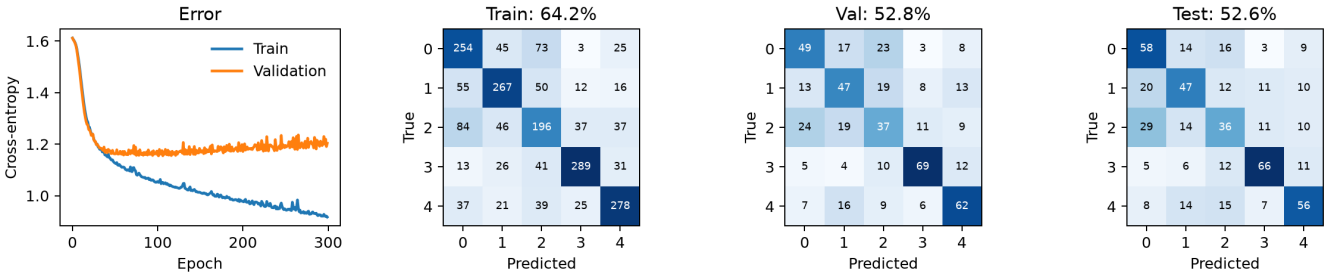


Result. Dropconnect at 5% gives the best validation accuracy. L2 with $\lambda = 1$ gives the best test accuracy. Strong L2 and batch normalization reduce generalization here.

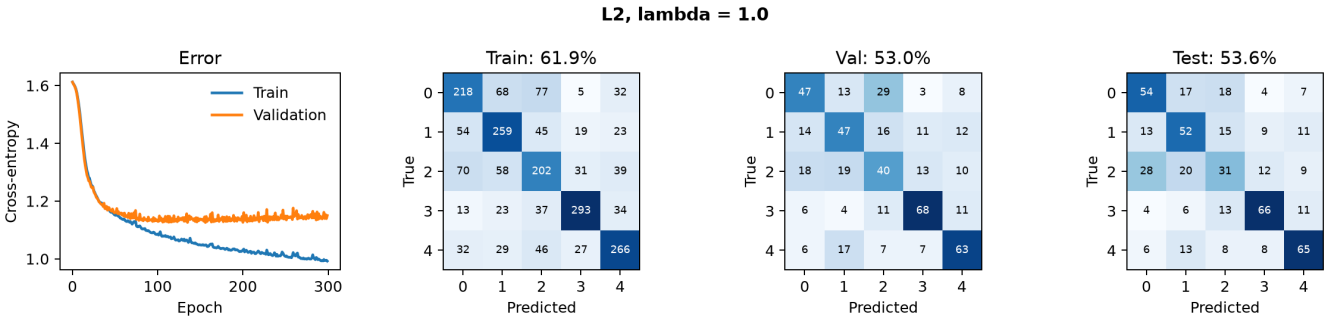
Error curves and confusion matrices



No regularization provides the reference result.

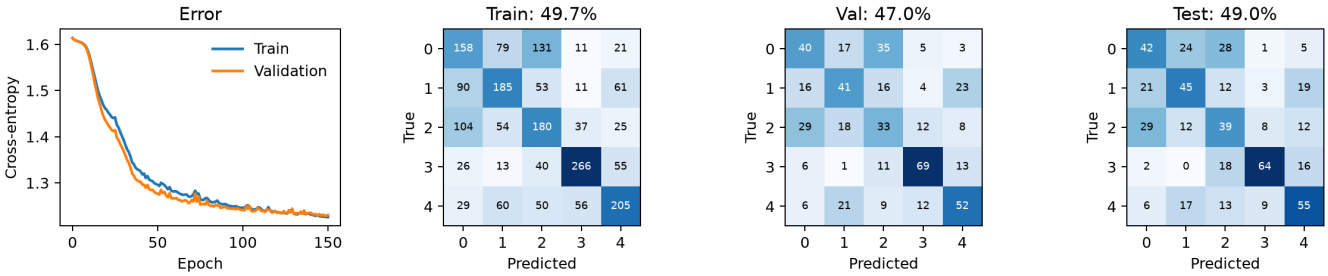


Light L2 gives similar test accuracy.



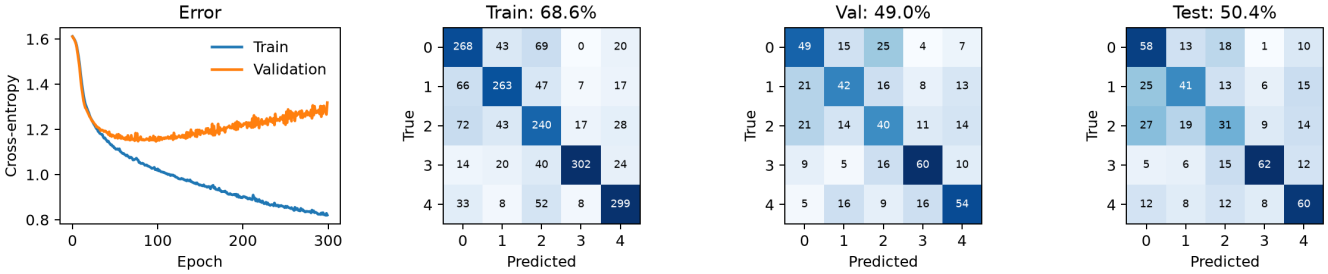
L2 with $\lambda = 1$ gives the highest test accuracy.

L2, lambda = 10.0



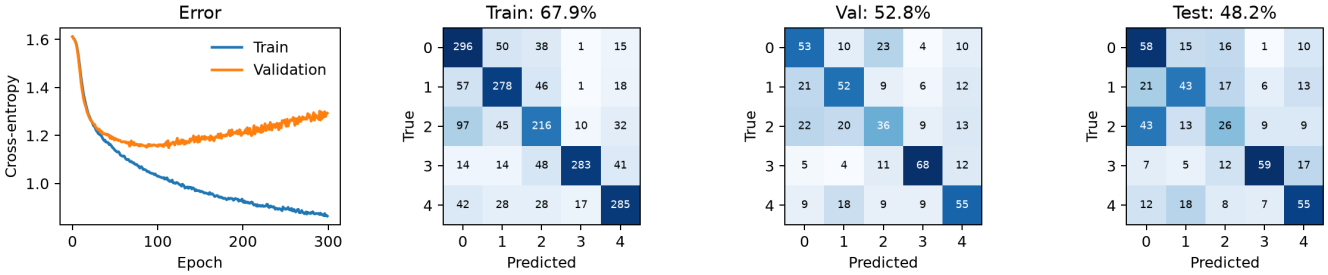
Strong L2 causes underfitting.

Dropout, 10%



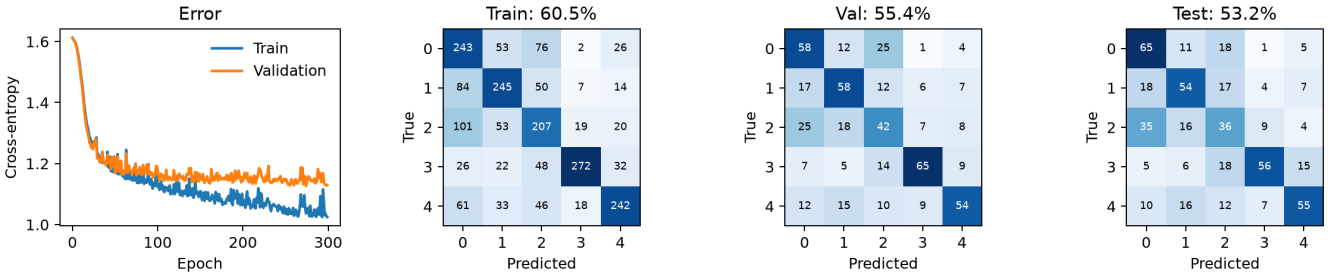
Dropout at 10% lowers validation accuracy.

Dropout, 20%



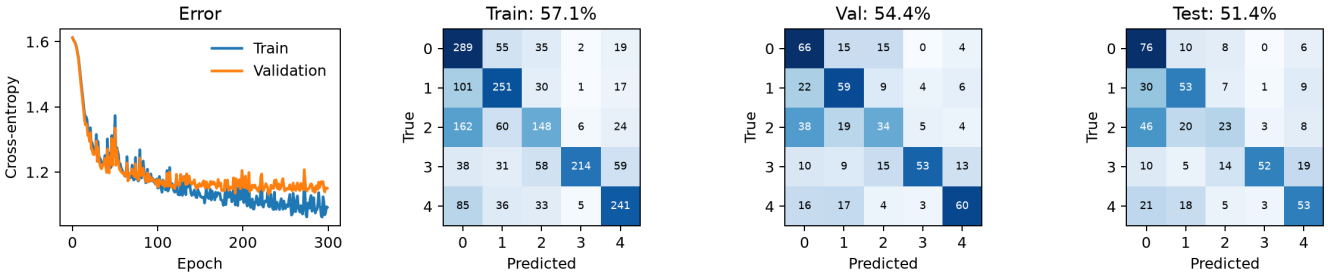
Dropout at 20% lowers test accuracy.

Dropconnect, 5%



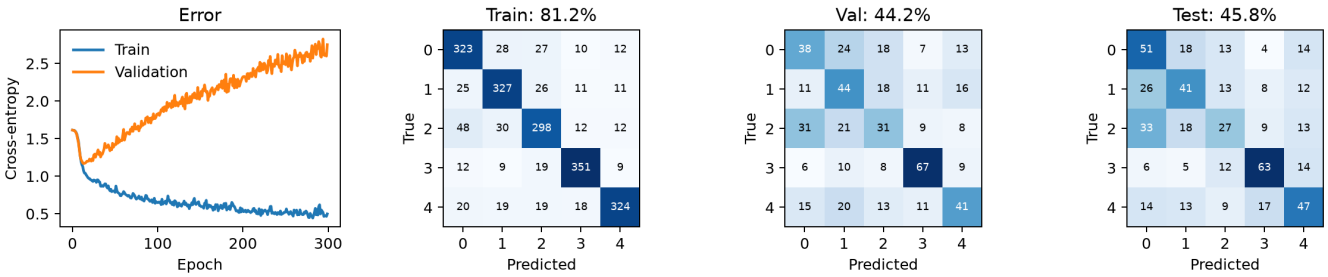
Dropconnect at 5% gives the highest validation accuracy.

Dropconnect, 10%



Dropconnect at 10% remains close to the baseline.

Post-activation batch norm



Batch normalization overfits this training split.